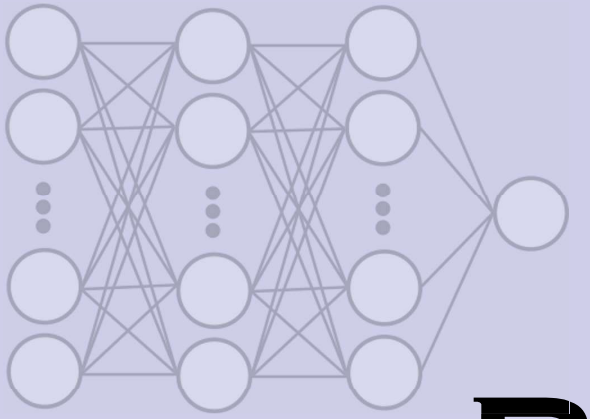
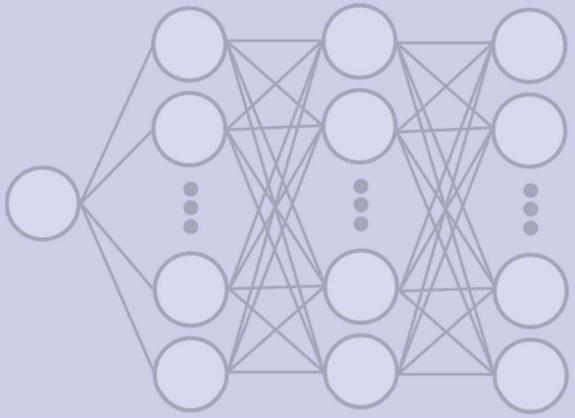


Projekt č.2

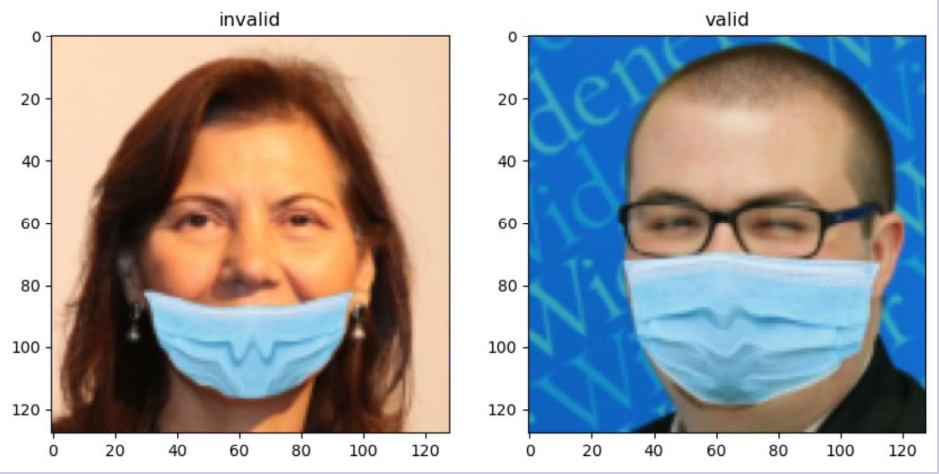
Hlboké neurónové siete



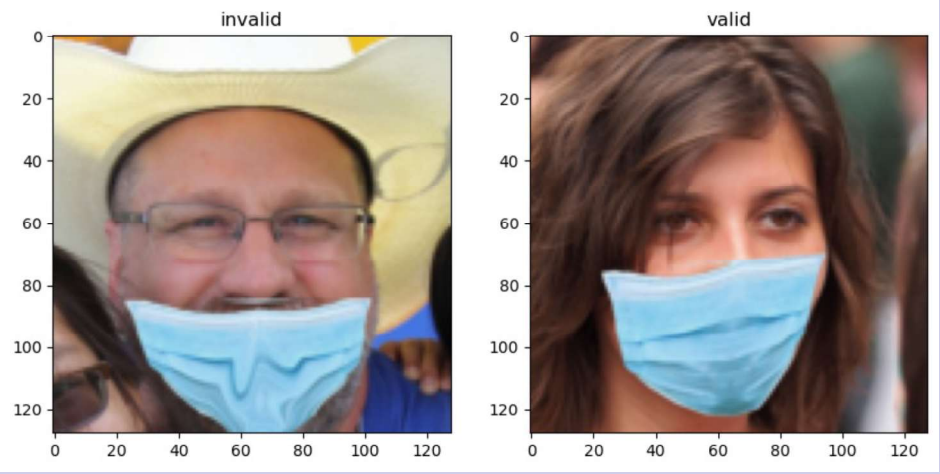
vypracovali: polc a trepáčová

zs 2024/2025

1. Rozpoznávanie dobre nasadeného rúška



ZLE NASADENÉ



DOBRE NASADENÉ

DATASET

POČET VZORIEK: 5 000

80% TRÉNOVACIE
20% TESTOVACIE










VLASTNÁ PARALELNÁ CNN

```
class ParallelCNN(nn.Module):
      def __init__(self, num_classes=2):
      super(ParallelCNN, self).__init__()
      # Branch 1
      self.branch1 = nn.Sequential(
            nn.Conv2d(3, 32, kernel_size=3, stride=1, padding=1), # Output: (32, 128, 128)
            nn.BatchNorm2d(32),
            nn.ReLU(),
            nn.MaxPool2d(kernel_size=2, stride=2), # Output: (32, 64, 64)

            nn.Conv2d(32, 64, kernel_size=3, stride=1, padding=1), # Output: (64, 64, 64)
            nn.BatchNorm2d(64),
            nn.ReLU(),
            nn.MaxPool2d(kernel_size=2, stride=2), # Output: (64, 32, 32)
            )
      # Branch 2
      self.branch2 = nn.Sequential(
            nn.Conv2d(3, 32, kernel_size=5, stride=1, padding=2), # Output: (32, 128, 128)
            nn.BatchNorm2d(32),
            nn.ReLU(),
            nn.MaxPool2d(kernel_size=2, stride=2), # Output: (32, 64, 64)

            nn.Conv2d(32, 64, kernel_size=5, stride=1, padding=2), # Output: (64, 64, 64)
            nn.BatchNorm2d(64),
            nn.ReLU(),
            nn.MaxPool2d(kernel_size=2, stride=2), # Output: (64, 32, 32)
            )
```

```
self.fc = nn.Sequential(
      nn.Linear(64*32*32*2, 512),
      nn.ReLU(),
      nn.Dropout(0.5),
      nn.Linear(512, num_classes)
      )


def forward(self, x):
      out1 = self.branch1(x)
      out2 = self.branch2(x)
      out1 = out1.view(out1.size(0), -1)
      out2 = out2.view(out2.size(0), -1)
      out = torch.cat((out1, out2), dim=1)
      out = self.fc(out)
      return out
```

BATCH SIZE: 256

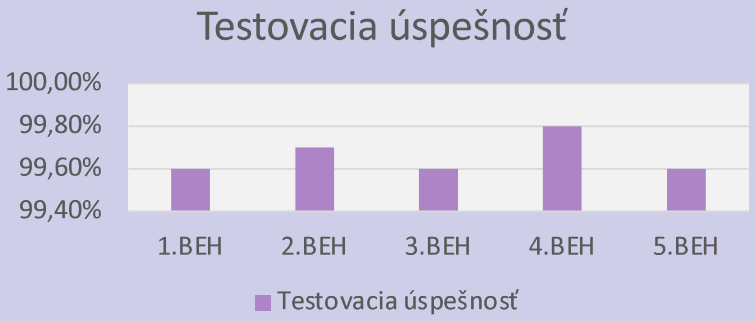
N_EPOCH: 20

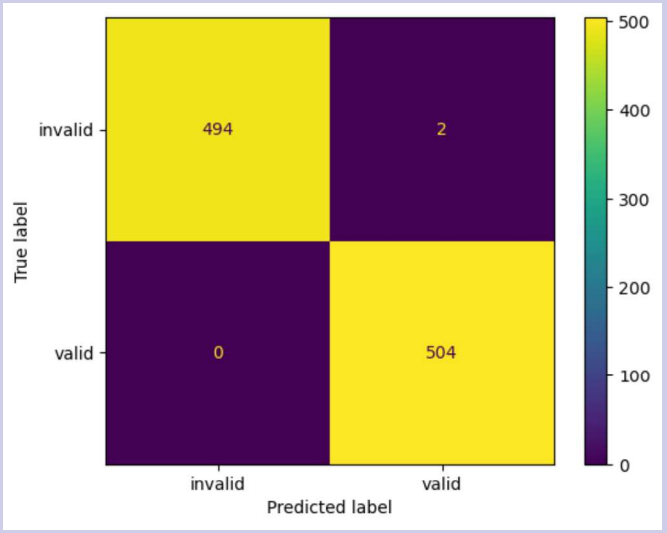
LEARNING RATE: 0.0004

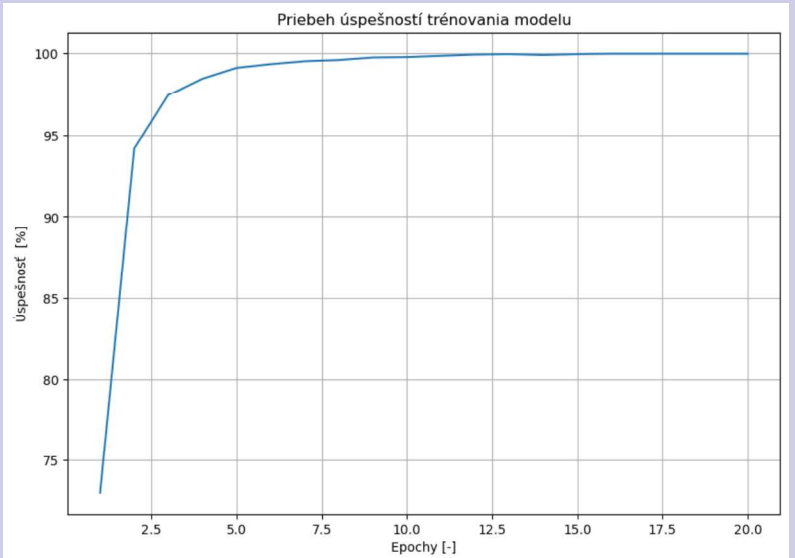
VÝSLEDKY

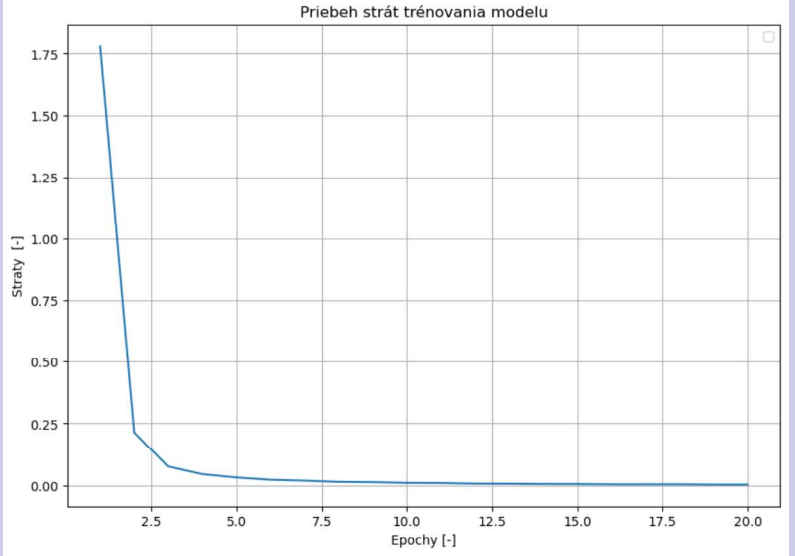
ÚSPEŠNOSŤ NA TESTOVACÍCH DÁTACH

1.BEH: 99.6%
2.BEH: 99.7%
3.BEH: 99.6%
4.BEH: 99.8%
5.BEH: 99.6%



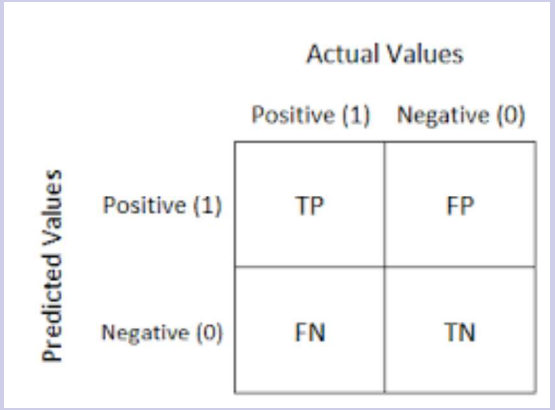






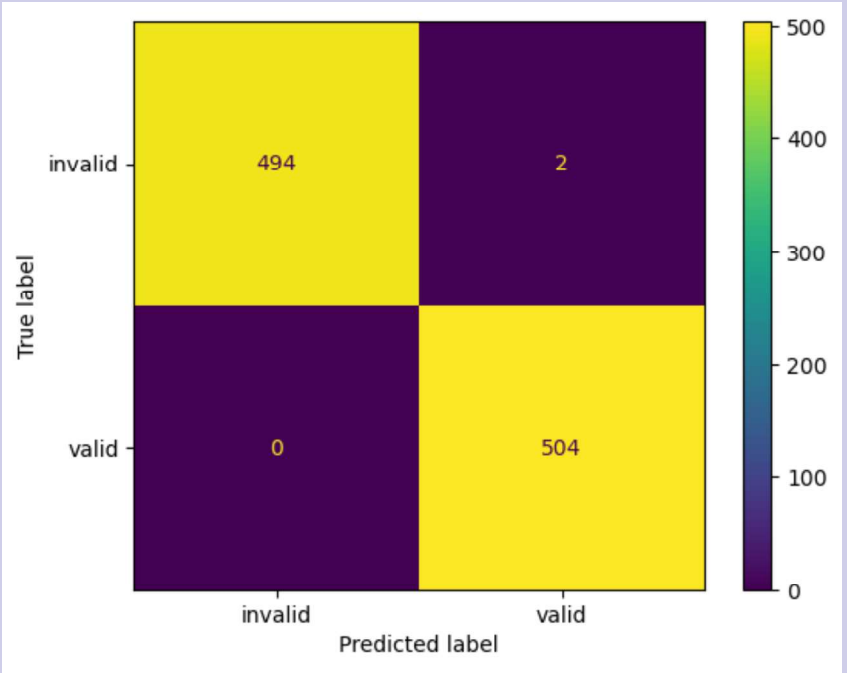
SENZITIVITA

$$\frac{True\ Positives\ (TP)}{True\ Positives\ (TP)+False\ Negatives\ (FN)} = 1$$



ŠPECIFICITA

$$\frac{True\ Negatives\ (TN)}{True\ Negatives\ (TN) + False\ Positives\ (FP)} = 0.996$$



BONUS – APLIKÁCIA CNN NA PC KAMERU

